

EE605 Quiz 3 Solutions

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1 Question 1

There can potentially be many solutions to this question, and some might be easier than the one given below. But this solution is straightforward. Let $C' = C_{\{q+1\}}$ i.e. the shortening of C at the last coordinate. Then clearly, a parity check matrix for C' is

$$\begin{pmatrix} 1 & 1 & \dots & 1 & 1 \\ \alpha_1 & \alpha_2 & \dots & \alpha_{q-1} & \alpha_q \\ \alpha_1^2 & \alpha_2^2 & \dots & \alpha_{q-1}^2 & \alpha_q^2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \alpha_1^{q-k} & \alpha_2^{q-k} & \dots & \alpha_{q-1}^{q-k} & \alpha_q^{q-k} \end{pmatrix}$$

where $\alpha_1, \dots, \alpha_{q-1}$ are the nonzero elements of $\mathbb{F}_q$ and α_q is the zero element of $\mathbb{F}_q$. Note then that a generator matrix for $C'^{\perp}$ is H' , and H' has order $q \times (q - k + 1)$ so $C'^{\perp}$ is a $(q, q - k + 1)$ -Reed-Solomon code $\Rightarrow C'^{\perp}$ is MDS $\Rightarrow C'$ is MDS $\Rightarrow d(C') = q - k + 2 \Rightarrow$ there is a set of $q - k + 2$ columns of H' which is linearly dependent, and every set of $q - k + 1$ columns of H' is linearly independent. So there is a set of $q - k + 2$ columns of H which is linearly dependent (since every column of H' is also a column of H). Now take a set of $q - k + 1$ columns from H . If all are also columns of H' , we are done. If not, the set must be some $q - k$ columns from H' and the column $(0 \ 0 \ \dots \ 0 \ 1)^T$. Make these columns into a square matrix. It will look like below.

$$\begin{pmatrix} 1 & 1 & \dots & 1 & 0 \\ \alpha_{i_1} & \alpha_{i_2} & \dots & \alpha_{i_{q-k}} & 0 \\ \alpha_{i_1}^2 & \alpha_{i_2}^2 & \dots & \alpha_{i_{q-k}}^2 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \alpha_{i_1}^{q-k-1} & \alpha_{i_2}^{q-k-1} & \dots & \alpha_{i_{q-k}}^{q-k-1} & 0 \\ \alpha_{i_1}^{q-k} & \alpha_{i_2}^{q-k} & \dots & \alpha_{i_{q-k}}^{q-k} & 1 \end{pmatrix}$$

where $i_1, i_2, \dots, i_{q-k}$ denote the indices of the chosen $q - k$ columns of H' . Note that this is a square matrix, so its columns are linearly independent iff its determinant is 0. Expanding the determinant along the last column, we get the value of its determinant to be

$$\det \left(\begin{pmatrix} 1 & 1 & \dots & 1 \\ \alpha_{i_1} & \alpha_{i_2} & \dots & \alpha_{i_{q-k}} \\ \alpha_{i_1}^2 & \alpha_{i_2}^2 & \dots & \alpha_{i_{q-k}}^2 \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{i_1}^{q-k-1} & \alpha_{i_2}^{q-k-1} & \dots & \alpha_{i_{q-k}}^{q-k-1} \end{pmatrix} \right)$$

which is a Vandermonde matrix. Since all of $\alpha_{i_1}, \alpha_{i_2}, \dots, \alpha_{i_{q-k}}$ are distinct, this determinant is nonzero. We are done.

Note that this Vandermonde matrix argument can also be used directly to show that if all the $q - k + 1$ columns of H are also columns of H' , then the determinant of the corresponding square submatrix is nonzero, so the columns must be linearly independent. That is arguably shorter than my solution. However, in that case, one needs to use the Singleton bound to conclude the exact value of d , because a priori we do not have a linearly dependent set of $q - k + 2$ columns.

2 Question 2

Given : C is an $[n, k, d]$ linear MDS code with generator matrix G , and G_J is a submatrix of G that comprises columns indexed by the elements of set J . For a received vector r , $\hat{m}_J = r_J G_J^{-1}$. The m^* that receives the maximum votes is chosen and decoded to codeword $\hat{c} = m^* G$. The number of errors is denoted by e .

2.a

For an $[n, k, d]$ linear MDS code C with generator matrix G , any set of k columns of G is linearly independent. Therefore, the true message m can be uniquely determined by taking any k columns of G , and the corresponding k columns of the received vector r , provided the received vector r is error-free.

Let G_J be a submatrix of G that comprises columns indexed by the elements of set J , where $J \subseteq \{1, \dots, n\}$. Let J_e be the set of e coordinates that are the error locations in the received vector r , and let m be the true message. If J comprises k distinct indices that do not overlap with J_e , then $m G_J = r_J$ and $m = r_J G_J^{-1}$. Note that since C is an MDS code, G_J is invertible.

Any k -sized J that does not overlap with J_e will contribute exactly one vote to m (the true message), resulting in a total of $\binom{n-e}{k}$ votes for m from such J . m can also receive votes from sets J of sizes $> k$. Thus the true message m will receive at least $\binom{n-e}{k}$ votes.

2.b

For any message $u \neq m$, there must be at least one error in r_J , i.e. J and J_e must overlap in at least one position, which is then used to decode to the incorrect message u . In such a case, the k columns in J used to get to the message u must come from the following :

- at least one column index from J_e ,
- any $\leq k - 1$ correct column indices from $\{1, \dots, n\} \setminus J_e$, since the number of indices of correct columns in J cannot exceed $k - 1$.

Therefore, the k columns used for decoding can be chosen in at most $\binom{e+k-1}{k}$ ways, and thus $u \neq m$ can receive at most $\binom{e+k-1}{k}$ votes.

2.c

The decoding is correct if the correct message m receives greater number of votes than any incorrect message $u \neq m$, i.e. $\binom{n-e}{k} > \binom{e+k-1}{k} \implies n - e > e + k - 1 \implies n - k + 1 > 2e$. Since C is an MDS code, $d = n - k + 1$ and $d > 2e$. Therefore, as long as $2e < d$, the decoding is correct.

2.d

Given : A linear code over $\mathbb{F}_7$ with generator matrix $G = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{bmatrix}$

2.d.i

The parameters $n = 4$, and $k = 2$ are directly obtained from G . Next, consider any $k = 2$ distinct columns $g_i = \begin{bmatrix} g_{i1} \\ g_{i2} \end{bmatrix} = \begin{bmatrix} 1 \\ g_{i2} \end{bmatrix}$ and $g_j = \begin{bmatrix} g_{j1} \\ g_{j2} \end{bmatrix} = \begin{bmatrix} 1 \\ g_{j2} \end{bmatrix}$ of the generator matrix G . g_i and g_j will be linearly independent if $\alpha g_i + \beta g_j = 0$ for some $\alpha, \beta \in \mathbb{F}_7$, where α and β are not both 0.

$\alpha g_i + \beta g_j = 0 \implies \alpha + \beta = 0, \alpha g_{i2} + \beta g_{j2} = 0$. Then, $\beta = -\alpha$ and $\alpha(g_{i2} - g_{j2}) = 0$. Since $g_{i2} \neq g_{j2}$, this implies that $\alpha = 0 \implies \beta = 0$. Thus any $k = 2$ columns of G are linearly independent and the code C is an MDS code.

2.d.ii

The received vector is $r = (2, 5, 1, 5)$, and contains exactly 1 error.

Decoding procedure : Take any $k = 2$ locations from r and obtain r_J . The corresponding $\hat{m}_J$ is obtained as $\hat{m}_J = r_J G_J^{-1}$, contributing to exactly one vote for the specific $\hat{m}_J$. The process is repeated for different k locations, until a message $m^* = [m_1^*, m_2^*]$ with $\binom{n-e}{k} = \binom{4-1}{2} = 3$ votes is obtained.

- $J = \{1, 2\}$: Let $u = [u_1, u_2]$ be the retrieved message. Then $[u_1 \ u_2] \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = [2 \ 5] \implies u_1 + u_2 = 2, u_1 + 2u_2 = 5 \implies u_2 = 3, u_1 = 6$, i.e. $u = [6, 3]$. Therefore, $\#votes([6, 3]) = 1$.
- $J = \{1, 3\}$: $[u_1 \ u_2] \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix} = [2 \ 1] \implies u_1 + u_2 = 2, u_1 + 3u_2 = 1 \implies u_2 = 3, u_1 = 6$, i.e. $u = [6, 3]$. Therefore, $\#votes([6, 3]) = 2$.
- $J = \{1, 4\}$: $[u_1 \ u_2] \begin{bmatrix} 1 & 1 \\ 1 & 4 \end{bmatrix} = [2 \ 5] \implies u_1 + u_2 = 2, u_1 + 4u_2 = 5 \implies u_2 = 1, u_1 = 1$, i.e. $u = [1, 1]$. Therefore, $\#votes([1, 1]) = 1$.
- $J = \{2, 3\}$: $[u_1 \ u_2] \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} = [5 \ 1] \implies u_1 + 2u_2 = 5, u_1 + 3u_2 = 1 \implies u_2 = 3, u_1 = 6$, i.e. $u = [6, 3]$. Therefore, $\#votes([6, 3]) = 3$.

Since $\#votes([6, 3]) = 3$, the decoded message is $m^* = [6, 3]$, and the corresponding decoded codeword is $\hat{c} = m^* G = [6 \ 3] \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{bmatrix} = [2, 5, 1, 4]$.

3 Question 3

3.a

3.a.i

Say we want to construct a polynomial $f(X) \in \mathbb{F}_q[X]$ of degree $n-1$ such that $f(x_i) = y_i$ for a prescribed set of n distinct points $x_i \in \mathbb{F}_q$ (so naturally $q \geq n$) and a prescribed set of n values $y_i \in \mathbb{F}_q$. If we can find n polynomials $f_i(X)$ for $i \in \{1, \dots, n\}$ such that $f_i(x_i) = 1$ and $f_i(x_j) = 0$ for $j \in \{1, \dots, n\} \setminus \{i\}$, then clearly,

$$f(X) = \sum_{i=1}^n y_i f_i(X)$$

Such polynomials f_i can be described very nicely. We have

$$f_i(X) = \prod_{j \in \{1, \dots, n\} \setminus \{i\}} \left(\frac{X - x_j}{x_i - x_j} \right)$$

It is easily checked that these f_i do indeed satisfy the property given above. Coming to the question, using the same notation,

$$f(X) = 2f_0(X) + 4f_1(X) + f_2(X)$$

where

$$f_0(X) = \left(\frac{X-1}{0-1} \right) \left(\frac{X-2}{0-2} \right) = 2^{-1}(X-1)(X-2) = 4(X^2 - 3X + 2) = 4X^2 + 2X + 1$$

$$f_1(X) = \left(\frac{X-0}{1-0} \right) \left(\frac{X-2}{1-2} \right) = (-1)(X)(X-2) = 6(X^2 - 2X) = 6X^2 + 2X$$

and

$$f_2(X) = \left(\frac{X-0}{2-0} \right) \left(\frac{X-1}{2-1} \right) = 2^{-1}(X)(X-1) = 4(X^2 - X) = 4X^2 + 3X$$

Of course, all arithmetic is done in $\mathbb{F}_7$. So we finally get

$$f(X) = 2(4X^2 + 2X + 1) + 4(6X^2 + 2X) + 4X^2 + 3X = X^2 + X + 2$$

and one can verify that $f(0) = 2, f(1) = 4$ and $f(2) = 1$.

3.a.ii

The codeword is $(2, 4, 1, f(3), f(4)) = (2, 4, 1, 0, 1)$.

3.b**3.b.i**

Each codeword in the given Reed-Solomon code is of the form $(f(0), f(1), f(2), f(3), f(4))$ where f is a polynomial with coefficients in $\mathbb{F}_7$ of degree at most 2. We are given here that $f(0) = 3$, $f(2) = 2$, and $f(3) = 6$. Again, using Lagrange interpolation, we get

$$f(X) = 3f_0(X) + 2f_2(X) + 6f_3(X)$$

We will again need to recompute the polynomials f_0 , f_2 and f_3 like in the previous part. They come out to be

$$f_0(X) = 6X^2 + 5X + 1$$

$$f_2(X) = 3X^2 + 5X$$

and

$$f_3(X) = 5X^2 + 4X$$

So the polynomial f is

$$f(X) = 3(6X^2 + 5X + 1) + 2(3X^2 + 5X) + 6(5X^2 + 4X) = 5X^2 + 3$$

Again, we can verify that this satisfies $f(0) = 3$, $f(2) = 2$ and $f(3) = 6$. So the original message was $(3, f(1), 2) = (3, 1, 2)$.

3.b.ii

The full codeword is clearly $(3, 1, 2, 6, f(4)) = (3, 1, 2, 6, 6)$.